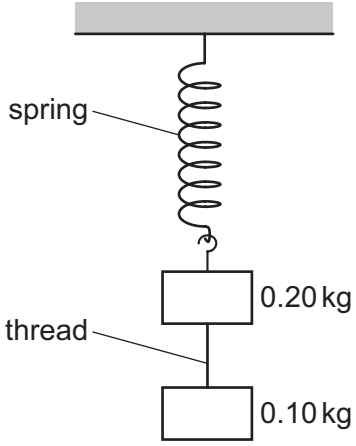
8 A mass of 0.20 kg is suspended from the lower end of a light spring. A second mass of 0.10 kg is suspended from the first mass by a thread. The arrangement is allowed to come into static equilibrium and then the thread is cut.



Immediately after the thread is cut, what is the upward acceleration of the 0.20 kg mass?

A $4.9\,m\,s^{-2}$ **B** $6.5\,m\,s^{-2}$ **C** $9.8\,m\,s^{-2}$ **D** $15\,m\,s^{-2}$